

Rafael Antonio Casella

Financial Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist focused on delivering analytical solutions for the financial and strategic sectors, combining a solid technical background in Computer Engineering with extensive experience in quantitative modeling and market analysis. Hands-on expertise developing end-to-end data pipelines (ETL), automating corporate processes using Python and SQL, and designing executive-level dashboards with Power BI and Streamlit. Specialized in leveraging quantitative models, Machine Learning algorithms, and predictive modeling to mitigate risk, optimize operational workflows, and generate business intelligence aimed at driving revenue growth and internal efficiency.

HARD SKILLS

Languages, Libraries & Web: Python (Pandas, NumPy, Matplotlib, Seaborn, Asyncio, Multithreading), HTML5, CSS3, JavaScript.

Databases & SQL: PostgreSQL, MySQL, Relational Queries, Data Extraction.

Machine Learning & I.A.: Scikit-Learn, TensorFlow, Keras, Logistic Regression, LSTM Neural Networks, Time Series, Feature Engineering.

Data Viz & Process Automation: Power BI, Data Storytelling, Streamlit, Playwright, Network Request Interception (API JSON), Robotic Process Automation (RPA).

Quantitative Finance: KPIs, Look-ahead Bias Mitigation, Technical-Statistical Market Analysis, Capital Risk Management, Volatility Indicators.

PROFESSIONAL EXPERIENCE

Financial & Data Professional (09/2025 – 05/2026)

Ceopag (Payments Fintech) Sao Jose do Rio Preto, SP, Brazil

- Developed SQL queries to automate executive reports, reducing decision-making time for the board of directors.
- Built a Logistic Regression model for a **Customer Churn Predictor**, achieving **82.15% accuracy** in early churn risk alerts.
- Executed complex bank reconciliations for accounts payable and receivable, ensuring the integrity of the audit database.

Market Analyst (Independent Trader) (02/2017 – 07/2025)

B3 (Brazilian Stock Exchange)

- Built an end-to-end **Quantitative Trading (Modular ETL)** infrastructure in Python, successfully mitigating look-ahead bias.
- Implemented a Recurrent Neural Network (**LSTM via TensorFlow/Keras**) for complex financial time series forecasting.
- Managed capital risk via Home Broker using quantitative algorithms grounded in intraday statistical patterns.

Co-Founder & Technical Analyst (06/2018 – 12/2021)

Portal do Mercado

- Created an automated pipeline using Playwright and Multithreading for a real-time **Geopolitical Intelligence Monitor** via JSON API.
- Led the technical management of the NysePro software, structuring data streaming pipelines for real-time news and financial metrics.
- Developed and optimized investment portfolios based on quantitative data analysis for community members.

DATA PROJECTS

CFO Strategic KPI Dashboard Development

Business Intelligence, Corporate Finance, Controlling, Data Viz

- Modeled and structured an executive-level dashboard designed to support high-level financial leadership (C-Level) decision-making in performance monitoring and liquidity risk mitigation.
- Mapped and engineered critical profitability and sales efficiency indicators, including Gross Profit Margin, Contribution Margin per Product, COGS, and Average Order Value.
- Implemented operational growth and scale metrics aimed at cash flow predictability, tracking Total Processed Volume (**TPV**) and Month-over-Month (**MoM**) growth rates.

Cohort Analysis & Banking Retention (LTV)

SQL, PostgreSQL, Power BI, Data Engineering

- Developed an end-to-end data pipeline focused on statistical and behavioral analysis of customer retention for financial institution.
- Structured and processed complex queries in **PostgreSQL** for null-value handling, data quality auditing, and automated time-interval calculation.
- Built Cohort matrices optimized directly at the database source, integrating the data model into Power BI for interactive visualization of customer decay curves and **Lifetime Value (LTV)** monitoring.

Credit Score Prediction & Risk Analysis

Python, Scikit-Learn, Linear Regression, Pandas, Machine Learning

- Developed a predictive Machine Learning model aimed at automating financial risk analysis and supporting credit concession decision-making.
- Conducted Exploratory Data Analysis (EDA) and outlier treatment on a database containing over 10,000 records of socioeconomic and asset profiles.
- Implemented and validated a Linear Regression algorithm via Scikit-Learn for continuous risk score forecasting, achieving a coefficient of determination (**R² Score**) of **79.5%**.

EDUCATION & LANGUAGES

Postgraduate Degree in Data Science, Machine Learning & AI

XP Educação (In Progress)

Bachelor of Science in Computer Engineering

UFSCar / UNORP (2013)

Português: Native | Inglês: Professional Reading & Comprehension